Drug Biomarker Prediction – Project Summary

Objective

The goal of this project was to predict the **BioMarkerValue** (a measure of drug response) based on the molecular structure of drugs represented using **SMILES** notation. This was accomplished using cheminformatics techniques and machine learning models.

Data & Features

- Input data includes:

In this project, I chose to use **Morgan Fingerprints** instead of the raw **SMILES** strings because Morgan Fingerprints represent the molecular structure in a fixed-length **numerical vector form**, making them more suitable for machine learning models. In contrast, SMILES are **text-based representations** that require complex parsing or embedding techniques to be usable in a predictive model.

Initially, I relied solely on Morgan Fingerprints as features. However, after further exploration of cheminformatics using the **RDKit** library, I discovered that incorporating **Molecular Descriptors** such as Molecular Weight, LogP, TPSA, HBD, and HBA could provide additional insights into the molecular properties. These descriptors added complementary information to the fingerprints, enhancing the model’s ability to learn relevant patterns. As a result, I included these descriptors alongside the fingerprints to improve the model's performance.

- `Drug_ID`: Unique identifier for the drug
- `Drug`: SMILES string representing the molecular structure

- `Bio_Marker_Value`: Target variable for training data

    - Features used:

1. **Morgan Fingerprints (2048-bit)** – captures molecular structure.
2. **Molecular Descriptors** (computed using RDKit):
    - Molecular Weight (MolWt)
    - LogP (Octanol-water partition coefficient)
    - TPSA (Topological Polar Surface Area)
    - HBD (Hydrogen Bond Donors)
    - HBA (Hydrogen Bond Acceptors)

Model Development Workflow

1. Model Exploration

Several models were evaluated to find the best fit for this problem:

- **Linear Regression**
  - Fast and interpretable but **overfit** the training data severely.
- **Support Vector Regressor (SVR)**
  - Provided better control over margin and generalization, but still showed **overfitting** and slow computation for high-dimensional features.
- **Random Forest Regressor**
  - Capable of modeling complex relationships but prone to **variance** and **overfitting** in this context.
- **XGBoost Regressor** (Final Choice)
  - Provided **the best generalization**.
  - Controlled overfitting effectively using regularization and boosting.
  - Outperformed all previous models in terms of **Mean Absolute Error (MAE)** and **R^2 Score**.

2. Training Process

- Data was split into training and test sets (80/20 split).
- Features (fingerprints + descriptors) were concatenated.
- Model was trained and evaluated on both sets.
- Final model was saved as `xgb_trained_model.json`.

Prediction Pipeline

- A script `valid_data_prediction.py` reads new data (e.g., `valid.csv`), computes fingerprints and descriptors, and generates predictions.
- Uses the saved XGBoost model.
- Outputs a file: `predicted_biomarker_values.csv`.

Output

- Final predictions for the validation data are saved in:
  - `submission.csv` – Contains: `Drug_ID`, `Drug`, and predicted `Bio_Marker_Value`

Conclusion

Through iterative experimentation and performance evaluation, **XGBoost** was chosen for its superior balance between bias and variance. By combining **chemical informatics** and **machine learning**, this project successfully predicts biomarker values from molecular data.